

DeRa: Deep-Range Low-Power Wide Area IoT via Ultra-sensitive Signal Detection

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Abstract

LoRa extends communication range by operating well below the noise floor, yet this capability hinges on reliable packet detection—an overlooked bottleneck. Carrier frequency offset (CFO) introduces deterministic inter-chirp phase shifts that prevent coherent preamble accumulation; because all existing detectors fall back on non-coherent, magnitude-only addition, they forfeit substantial signal power. We present DeRa, a phase-coherent LoRa detection and decoding framework built on three contributions: (1) a phase-coherent detection method that exploits the relationship between CFO and inter-chirp phase evolution to detect weak packets; (2) analytical compensation of compound CFO-induced scaling distortions—timing drift, frequency migration, and energy leakage—so that coherent accumulation remains effective even under 40 kHz CFO, eliminating the need for expensive oscillator hardware; and (3) a CRC-guided, phase-compensated iterative decoder that coherently recombines split symbol energy and closes the detection-to-decoding feedback loop. On COTS LoRa devices across SF 7–12, DeRa lowers the detection threshold by 4.3–4.8 dB and the decoding threshold by 3.3–3.9 dB over state-of-the-art baselines, while maintaining $O(N \log N)$ per-symbol complexity and consistent gains in real-world field experiments.

CCS Concepts

• **Networks** → **Network protocols**; **Network architectures**.

Keywords

Internet of Things, low-power wide-area networks, LoRa, weak-signal detection

ACM Reference Format:

Jialuo Du, Jingkai Lin, and Zhichao Cao. 2026. DeRa: Deep-Range Low-Power Wide Area IoT via Ultra-sensitive Signal Detection. In *The 32nd Annual International Conference on Mobile Computing and Networking (MobiCom '26)*, October 26–30, 2026, Austin, TX, USA. ACM, New York, NY, USA, 21 pages. <https://doi.org/10.1145/3795866.3844157>

1 Introduction

LoRa (short for Long Range) [33] is designed for long-range Internet of Things (IoT) communication at low signal-to-noise ratios (SNRs) [3, 4, 9, 18, 20, 27, 30, 32, 36, 37, 39, 53]. Consequently, improving receiver sensitivity to weak LoRa signals is a central research problem.

Research on this problem, however, has focused heavily on weak-signal decoding [11, 12, 28, 40]. In contrast, weak-signal detection, the crucial initial step of identifying a packet's presence, remains a largely overlooked bottleneck (Figure 1). This oversight is critical because detection acts as the absolute gatekeeper. Regardless of how sophisticated a decoder might be, it cannot recover a packet if the receiver fails to detect it initially. In fact, some studies view packet detection as the primary performance bottleneck in practice [35].

LoRa detection relies on the principle of energy condensation [11, 28]. The process begins by spreading signal energy across a long preamble composed of multiple chirps; during detection, the receiver dechirps these preamble chirps, collapsing the distributed energy into a single, high-magnitude spectral peak. This concentration is what allows the signal to be pulled from beneath the noise floor. Ideally, a signal without phase shifts allows for coherent detection, where energy accumulates into a high-power peak, as illustrated in Figure 1(a). In this scenario, the alignment of the phase across symbols ensures that the integration of the signal significantly outweighs the stochastic noise.

However, real-world detection remains fragile because CFO induces a deterministic inter-chirp phase rotation between consecutive dechirped preamble chirps. As shown in Figure 1(b), where red triangles mark abrupt phase shifts, these distortions prevent the coherent energy accumulation needed for reliable detection. Because the phases do not align,



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MobiCom '26, Austin, TX, USA

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ACM ISBN 979-8-4007-2505-0/26/10

<https://doi.org/10.1145/3795866.3844157>

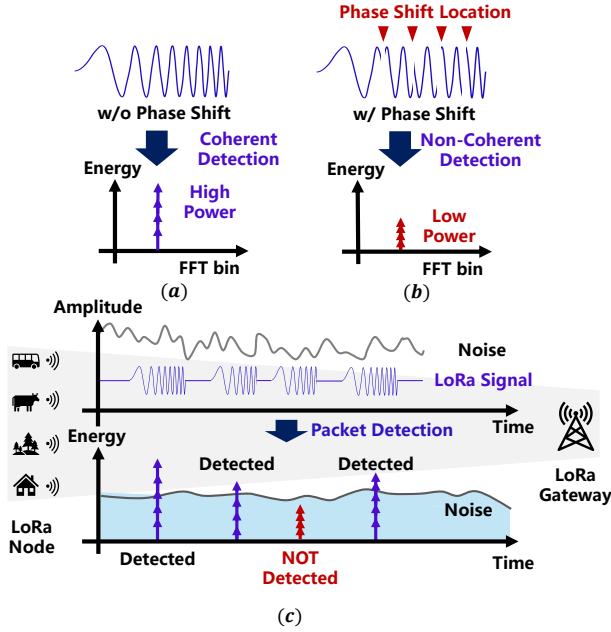


Figure 1: Packet Detection.

the peaks “cancel out” or fail to stack effectively, leaving the signal buried beneath the noise floor and leading to detection failure (Figure 1(c)). Despite various attempts to improve detection—multi-gateway combining in CloudLoRa [46], energy condensation via chirp trimming in LoRaTrimmer [12], and parabolic phase modeling in NELoRa [28]—all existing detectors ultimately fall back on non-coherent, magnitude-based preamble accumulation that discards phase information entirely, losing significant signal energy in the process.

To address this gap, we exploit the deterministic relationship between CFO and inter-chirp phase evolution to enable phase-coherent preamble accumulation. We present DeRa, a phase-coherent LoRa detection and decoding framework that exploits this CFO-phase structure. By modeling the relationship between the CFO and the resulting inter-chirp phase pattern, DeRa performs joint detection and CFO estimation. This dual capability simultaneously locates the packet and compensates for the CFO, which restores coherent energy accumulation and improves detection sensitivity at low SNRs. The model also extends to the payload to enable phase-compensated decoding, effectively closing the feedback loop between detection and decoding through CRC verification. However, translating this approach into a practical system requires addressing three key challenges.

Challenge 1: The CFO–Phase Chicken-and-Egg Problem. Coherent preamble accumulation demands that phase shifts be compensated before summation, yet these shifts are entirely governed by the unknown CFO—a dynamic value the detector seeks to estimate. This creates a fundamental circular dependency: accurate phase alignment requires a

precise CFO estimate, and a precise CFO estimate requires coherent energy accumulation, which in turn requires accurate phase alignment. Worse still, at the low SNRs where coherent gain is most needed, per-symbol phase measurements are heavily corrupted by noise and direct frequency estimation loses resolution, rendering both the CFO and the resulting phase trajectory practically unobservable through conventional means.

Challenge 2: Compound Scaling Distortions under Extreme CFO. CFO does not merely shift the carrier frequency; it simultaneously scales the symbol duration, bandwidth, and chirp rate of every transmitted chirp. These compound scaling effects cause cumulative timing drift and progressive frequency migration across symbol windows, pulling spectral peaks out of their expected FFT bins and destroying the alignment that detection relies upon. Low-cost on-chip oscillators can generate CFOs of up to 40 kHz, a regime in which all existing detectors fail entirely [12, 46]. While high-precision Temperature Compensated Crystal Oscillators (TCXOs) can suppress CFO to ~ 50 Hz, they impose prohibitive cost and power penalties on end nodes, negating the economic and energy advantages that define LPWANs. A practical detector must therefore embrace—not avoid—the full range of realistic CFOs and analytically compensate for their compound effects.

Challenge 3: End-to-End Phase Contamination in Payload Decoding. The same CFO that disrupts preamble accumulation contaminates every payload symbol: it introduces intra-symbol phase discontinuities at chirp wrap points and inter-symbol phase rotations across the entire packet. These distortions split dechirped energy across multiple FFT bins and misalign phase across symbols, destroying the tight spectral concentration that LoRa decoding depends on. Existing decoders either resort to magnitude-only processing [12, 46, 68] or rely on noise-sensitive features [11, 28, 40], both of which forfeit the ~ 3 dB coherent gain available from correct phase combination—a critical margin at low SNR.

DeRa addresses all three challenges through a unified, CFO-centric framework.

Breaking the chicken-and-egg loop via coherent accumulation. DeRa detects packets using a two-stage FFT. It first applies an FFT to condense signal power in each chirp, and then performs a column-wise FFT across chirps to coherently accumulate power while compensating for the CFO-induced inter-chirp phase rotation. In this way, we sidestep the circular dependency altogether: only the *correct* CFO candidate produces constructive phase alignment across all preamble symbols, generating a sharp power peak that rises decisively above the noise floor. All other candidates yield destructive or random phase combinations that remain buried in noise. This transforms CFO estimation from a fragile frequency measurement into a robust coherent-power peak

detection problem, simultaneously resolving both packet detection and CFO estimation in a single, noise-resilient operation.

Searching, compensating, and accumulating under extreme CFO. Beyond the phase rotation addressed above, CFO induces a cascade of additional distortions—timing drift, frequency migration, and energy leakage—that destroy the spectral alignment detection depends on. Based on the CFO estimate and spectrogram produced by the two-stage FFT, DeRa uses the closed-form model (Section 3) to predict the resulting phase rotations and align the corresponding spectral components coherently. These compensations remain effective at CFOs of up to 40 kHz, eliminating the need for high-precision oscillator hardware.

CRC-guided, phase-compensated iterative payload decoding. Leveraging the preamble-derived CFO estimate, DeRa analytically predicts and compensates the intra-symbol phase discontinuities using the closed-form model, coherently recombining split energy of payload symbols that magnitude-based decoders discard. When the initial decode fails CRC verification—signaling residual CFO estimation error—DeRa does not simply discard the packet. Instead, it performs a small, fixed number of retry trials, reusing the previous FFT results to evaluate the next-best CFO candidate, re-align the per-symbol phases, and re-decode until CRC passes or the retry budget is exhausted. This strategy couples detection and decoding by using payload CRC feedback to guide CFO refinement, thereby closing the loop between the two stages.

We implement and evaluate DeRa on commercial off-the-shelf (COTS) LoRa devices across spreading factors SF 7–12. Compared to state-of-the-art baselines including CloudLoRa [46] and LoRaTrimmer [12], DeRa lowers the detection threshold by **4.3–4.8 dB** and the decoding threshold by **3.3–3.9 dB**, while remaining robust under arbitrarily large CFO, tested up to 40 kHz on Semtech SX1280 nodes [43]. Based on a prior LoRa measurement study [39], these sensitivity improvements translate into estimated gains of 40.7% in node battery life and 47.4% in coverage area. DeRa maintains $O(N \log N)$ complexity—the same asymptotic complexity as standard FFT-based detection—and operates end-to-end in real time on a Raspberry Pi. The main contributions of this paper are as follows:

- We design a phase-coherent detection method that exploits the deterministic relationship between CFO and inter-chirp phase evolution at low SNR. A two-stage FFT first condenses signal power within each chirp and then coherently accumulates power across chirps, producing a noise-resilient peak at the correct CFO and thereby jointly performing packet detection and CFO estimation.

- We analytically model and compensate all compound CFO-induced distortions—including timing drift, frequency migration, and energy leakage—enabling coherent energy accumulation for robust detection even under 40 kHz CFO without expensive oscillator hardware. For decoding, DeRa employs a CRC-guided, phase-compensated iterative strategy that coherently recombines split symbol energy discarded by magnitude-based decoders, closing the detection-to-decoding feedback loop.
- We evaluate DeRa on COTS LoRa devices across SF 7–12. Results show 4.3–4.8 dB and 3.3–3.9 dB reductions in the detection and decoding thresholds, respectively, over state-of-the-art baselines, with robust operation under up to 40 kHz CFO confirmed in real-world field experiments.

2 Background and Motivation

2.1 LoRa Detection via Energy Condensation

LoRa employs Chirp Spread Spectrum (CSS) modulation [8, 54], where each symbol is a frequency chirp sweeping linearly across bandwidth BW over symbol duration $T = 2^{SF}/BW$, with SF denoting the spreading factor. At the receiver, *dechirping*—multiplying by a conjugate reference chirp—condenses the spread energy into a single sharp FFT peak, enabling signal recovery well below the noise floor.

A LoRa packet begins with a *preamble* of N_{pre} identical upchirps (typically 8), followed by sync-word, SFD, and payload symbols. Because preamble chirps are identical, their dechirped FFT outputs ideally produce peaks at the same bin with consistent phase. The receiver accumulates these outputs across all preamble windows, further concentrating energy. This two-level condensation—within each chirp (dechirp-FFT) and across chirps (accumulation)—is the mechanism that enables LoRa to detect signals at SNRs as low as -20 dB.

2.2 CFO Disrupts Coherent Accumulation

In practice, transmitter and receiver oscillators are not perfectly matched, introducing a carrier frequency offset (CFO). CFO causes two intertwined effects on the dechirped preamble outputs: (1) a *progressive frequency drift* that shifts the FFT peak across bins from symbol to symbol, and (2) a *progressive phase rotation* that destroys phase consistency across symbols. Because CFO is relatively stable over the duration of a packet, DeRa models it as approximately constant. Under this assumption, both effects grow linearly with symbol index, as CFO slightly scales the true symbol duration and causes cumulative timing misalignment (formalized in Section 3).

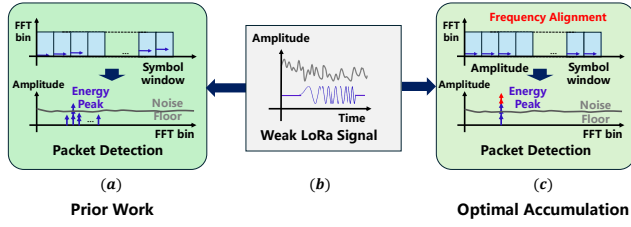


Figure 2: Weak LoRa packet detection: prior work vs. the optimal accumulation method.

Real-life experiments at SF = 12 and BW = 125 kHz quantify the severity. Even a CFO of just ± 4 Hz produces ~ 10 rad of unwrapped phase across 8 preamble symbols, which is enough to destroy coherent addition. In practice, CFOs can reach 40 kHz (as observed on Semtech SX1280 [43]), rendering phase-aligned accumulation entirely hopeless without compensation.

2.3 Motivation

Because CFO destroys inter-symbol phase consistency, existing detectors abandon phase information and fall back to *non-coherent* accumulation—summing magnitudes or powers (Figure 2(a)) [12, 46, 68]. While this sidesteps the phase problem, noise accumulates alongside the signal, yielding only $5 \log_{10}(N_{pre})$ dB of SNR improvement versus $10 \log_{10}(N_{pre})$ dB for coherent addition—a ~ 4.5 dB sensitivity gap at $N_{pre} = 8$ (see Appendix A.6 for a formal derivation). The situation worsens under large CFO: beyond phase rotation, compound scaling distortions—timing drift and progressive frequency migration—pull spectral peaks out of their expected FFT bins from symbol to symbol (Figure 2(b)), so even magnitude-based summation can no longer align power across symbols, causing prior detectors to degrade severely or fail entirely.

The inefficiency motivates us to analytically compensate both the inter-symbol phase trajectory and these compound frequency distortions, ensuring that energy accumulates coherently and accurately into sharp, high-power peaks that rise decisively above the noise floor, as shown in Figure 2(c).

3 Signal Model: CFO-Induced Phase Distortion

Carrier frequency offset (CFO) is traditionally treated as a nuisance that distorts frequency measurements and degrades detection. CFO-induced phase evolution is a known property of CSS signals [1, 60]. Here, we express the frequency and phase of each FFT peak as closed-form functions of the CFO f_{cfo} and the initial timing offset (TO) τ_0 , enabling DeRa to compensate for this evolution and coherently accumulate signals at low SNR. The resulting model supports the efficient detection and decoding algorithm in Section 4.

3.1 Three Scaling Effects of CFO

Let f_c denote the nominal carrier frequency and $\epsilon = f_{cfo}/f_c$ the relative frequency offset between transmitter and receiver oscillators. Apart from a uniform frequency shift of f_{cfo} across the whole signal, CFO also scales the transmitted waveform in three intertwined ways:

- **Duration scaling:** The actual symbol duration becomes $T' = T(1 - \epsilon)$.
- **Bandwidth scaling:** The chirp sweep range becomes $BW' = BW(1 + \epsilon)$.
- **Chirp-rate scaling:** The nominal chirp rate $\beta = BW/T$ changes to $\beta' = BW'/T' = \beta(1 + \epsilon)/(1 - \epsilon) \approx \beta(1 + 2\epsilon)$.

3.2 From Scaling to Phase: The Dechirped Residual

Figure 3 traces how the three scaling effects propagate through the receiver's dechirp-FFT pipeline.

(a) Ideal case (Figure 3(a)). When $\epsilon = 0$, the received upchirp is perfectly matched by the local downchirp reference. Dechirping yields a flat frequency at $f = 0$, all energy concentrates in FFT bin 0, and the dechirped phase remains identically zero.

(b) Under CFO (Figure 3(b)). The received chirp sweeps at rate β' over bandwidth BW' , while the receiver's reference uses the nominal β and BW . Dechirping—multiplying the received signal by the conjugate reference—leaves a residual with instantaneous frequency $f_{dc}(t)$, where t is local time within the symbol:

$$f_{dc}(t) = \underbrace{f_{cfo}}_{\text{start freq}} - \underbrace{\frac{BW\epsilon}{2}}_{\text{rate mismatch}} + (\beta' - \beta) \cdot t \quad (1)$$

The start frequency is dominated by f_{cfo} ; the rate mismatch $\beta' - \beta \approx 2\beta\epsilon$ introduces a slow linear drift. Because the FFT coherently sums this evolving phase across the entire symbol, the peak carries a net phase offset reflecting the cumulative distortion.

Meanwhile, duration scaling produces the accumulated TO τ_i at symbol index i :

$$\tau_i = \tau_0 - i\epsilon T \quad (2)$$

When $T' < T$, each transmitted symbol finishes slightly earlier than the receiver expects, producing the cumulative timing drift $-i\epsilon T$; when $T' > T$, the drift reverses direction.

(c) Accumulated offset compounds both effects (Figure 3(c)). By the i -th symbol, the time offset τ_i has shifted the chirp boundary within the receiver's window. This shifts the dechirped frequency:

$$f_i(t) = f_{dc}(t) + \beta' \tau_i \quad (3)$$

causing the FFT peak to drift progressively across bins. It also means the chirp has already evolved for duration $|\tau_i|$

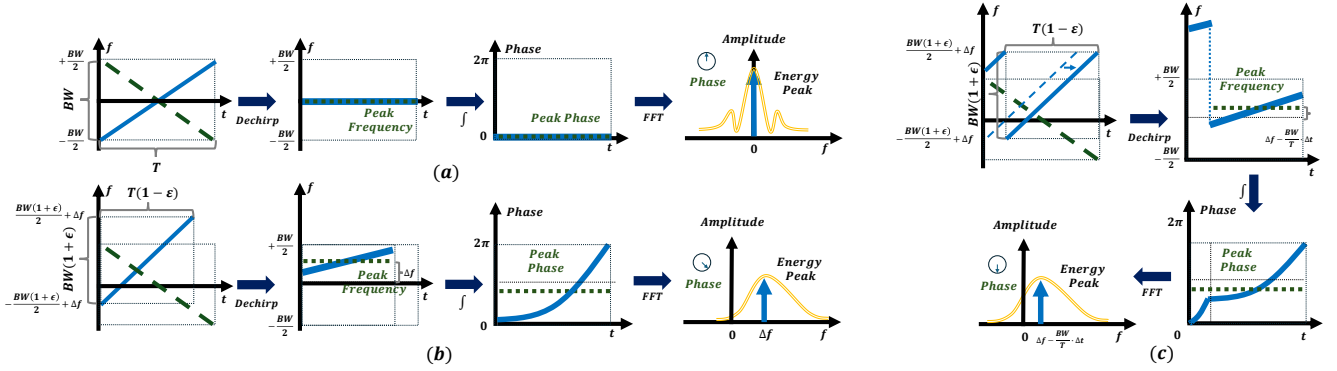


Figure 3: Impact of CFO on the dechirp-FFT pipeline. (a) Ideal case: dechirping yields a pure DC tone and zero phase. (b) CFO introduces residual frequency drift and phase curvature; duration scaling causes cumulative time offset. (c) Accumulated time offset shifts the dechirped frequency and adds a growing initial phase to each symbol.

before the receiver's window begins, accumulating an initial phase:

$$\phi_i^{(0)} = 2\pi \left[(f_{cfo} - \frac{BW\epsilon}{2})|\tau_i| + \frac{\beta' - \beta}{2}|\tau_i|^2 \right] \quad (4)$$

The dechirped residual phase. Using local time $t' = t - iT$ within the i -th receiver window, let $N = 2^{SF}$ denote the number of FFT bins per symbol and $c \in \{0, \dots, N-1\}$ the integer cyclic shift ($c = 0$ for preamble). The full dechirped residual phase for chirp direction v (+1 for upchirps and -1 for downchirps) is:

$$\begin{aligned} \Phi_{\text{res}}(t') = & \underbrace{\pi(\beta' - \beta)v}_{\text{chirp-rate mismatch}} t'^2 \\ & + 2\pi \underbrace{(f_{cfo} + BW'(\frac{c}{N} - \frac{v}{2}) + \beta'\tau_i v + \frac{BW \cdot v}{2})}_{\text{dechirped center frequency}} t' + \underbrace{\phi_i}_{\text{frame phase}} \end{aligned} \quad (5)$$

In practice, stochastic phase noise and within-packet CFO variation appear as an additional residual phase term $\eta_i(t')$ on top of $\Phi_{\text{res}}(t')$. DeRa assumes that this residual remains sufficiently small over the preamble accumulation interval and has minimal effect on the coherent-power peak.

The three terms have distinct physical roles:

- **Quadratic term** ($\pi(\beta' - \beta)v \cdot t'^2$): Arises from the chirp-rate mismatch between the received and reference chirps. Its magnitude is bounded by $2\pi\epsilon \cdot 2^{SF} \approx 0.44$ rad even in the worst case (SF12 at 40 kHz CFO; see Appendix A.5). In the *frequency* domain, this causes slight peak broadening that is negligible—nearly all energy stays in a single FFT bin. In the *phase* domain, its contribution is absorbed as a deterministic constant by the FFT (discussed in Appendix A.5).
- **Linear term:** Determines *which FFT bin* the energy falls into. For preamble upchirps ($c = 0, v = +1$), the

center frequency reduces to approximately $f_{cfo} + \beta'\tau_i$, encoding both CFO and time offset.

- **Constant term** (ϕ_i): The accumulated frame phase at symbol i . Under the phase-continuity constraint that threads a single unbroken phase trajectory across the entire frame (see Appendix A), the phase increment per symbol is uniform:

$$\Delta\phi = 2\pi f_{cfo}T' \pmod{2\pi} \quad (6)$$

regardless of symbol type or code value, because all terms involving $BW' \cdot T' = 2^{SF}$ contribute integer multiples of 2π (Appendix A). Letting ϕ_0 denote the starting phase of the packet, the frame phase is:

$$\phi_i = \phi_0 + 2\pi f_{cfo} \cdot i \cdot T' \quad (7)$$

This is the key quantity for coherent combination.

Since the quadratic phase is small, the dechirped signal is well approximated as a single tone. For N_s samples per symbol, the FFT output $Y_i[k^*]$ at peak bin k^* therefore carries two pieces of information:

$$Y_i[k^*] \approx N_s \cdot e^{j\phi_i}, \quad j = \sqrt{-1} \quad (8)$$

- **Bin position** k^* : encodes $f_{cfo} + \beta\tau_i$ for upchirps, $f_{cfo} - \beta\tau_i$ for downchirps—the TO contribution reverses sign.
- **Complex angle** $\angle Y_i[k^*]$: encodes ϕ_i , the accumulated frame phase.

The total phase at the FFT peak combines the frame phase with a small contribution from the pre-window evolution and fractional-bin scalloping:

$$\angle Y_i[k^*] \approx \phi_0 + 2\pi f_{cfo} \cdot i \cdot T' + \phi_i^{(0)} + \psi(\delta_i) \quad (9)$$

where $\psi(\delta_i)$ is a deterministic function of the fractional-bin offset (derived in Appendix A.5). The dominant contribution is the linearly increasing frame phase $2\pi f_{cfo} \cdot i \cdot T'$.

3.3 Two-Section Windowing Model

A subtlety arises because the receiver segments the signal into windows of nominal duration T , while the actual symbol duration is $T' \neq T$. As a result, each receiver window does not align with a single transmitted symbol but instead captures *two sections* of the transmitted signal (Figure 3(c)):

- **Section 1 (Sub-1):** The tail of the current chirp symbol, spanning from τ_i to T within the window.
- **Section 2 (Sub-2):** The head of the next chirp symbol, spanning from 0 to τ_{i+1} within the window.

The relative lengths of Sub-1 and Sub-2 are determined by the timing offset. Because the initial signal arrival time is effectively random relative to the receiver window, the two sections have equal expected duration and are therefore each expected to contain half of the signal energy under a uniform timing-offset assumption. Consequently, DeRa includes both contributions when forming the coherent detection score.

Each section produces an independent complex contribution at a distinct frequency in the dechirped FFT. We parameterize each section by its center frequency F and phase P :

- (F_1, P_1) : the frequency and phase of Section 1.
- (F_2, P_2) : the frequency and phase of Section 2.

Both pairs are fully determined by (f_{cfo}, τ_0, i, c) . The exact expressions, derived from evaluating Eq. 5 over each section's time interval (see Appendix A.4), are:

Section 1 (current symbol, spanning $[\tau_i, T]$ in the receiver window):

$$F_1 = \frac{2\pi(\beta' - \beta)v \cdot t_{\text{mid},1} + B_1}{2\pi}, \quad t_{\text{mid},1} = \frac{\tau_i + T}{2} \quad (10)$$

$$P_1 = \underbrace{2\pi f_{cfo} T' \cdot i + \pi \beta' v \tau_i^2}_{\text{frame phase}} + 2\pi \tau_i \left(\frac{BW'v}{2} - f_{cfo} - f_{\text{mod}} \right) - \pi(\beta' - \beta)v \cdot t_{\text{mid},1}^2$$

where $f_{\text{mod}} = (c/N) \cdot BW'$ is the codeword-induced frequency shift and $B_1 = 2\pi(f_{cfo} - \frac{BW'v}{2}v + f_{\text{mod}} - \beta'v\tau_i)$ is the linear coefficient.

Section 2 (next symbol, spanning $[0, \tau_{i+1}]$, if $\tau_{i+1} > 0$):

$$F_2 = \frac{2\pi(\beta' - \beta)v \cdot t_{\text{mid},2} + B_2}{2\pi}, \quad t_{\text{mid},2} = \frac{\tau_{i+1}}{2} \quad (11)$$

$$P_2 = 2\pi f_{cfo} T' \cdot i + \pi \beta' v (\tau_i - T)^2 - 2\pi(\tau_i - T) \left(-\frac{BW'v}{2} + f_{cfo} + f_{\text{mod}} \right) - \pi(\beta' - \beta)v \cdot t_{\text{mid},2}^2$$

where $B_2 = 2\pi(f_{cfo} - \frac{BW'v}{2}v + f_{\text{mod}} - \beta'v(\tau_i - T))$ is the corresponding linear coefficient.

Fractional-bin phase correction. In the FFT, F_1 (or F_2) will generally not land exactly on an integer bin. The fractional-bin offset $\delta = F - f_{\text{bin}}$, where f_{bin} is the nearest FFT-bin frequency, produces a deterministic scalloping phase shift

$2\pi \cdot \delta \cdot t_{\text{mid}}$ that is small in *frequency* (sub-bin) but can be significant in *phase* (several radians over many symbols). This sub-bin phase correction must be accounted for when coherently combining symbols; the algorithm of Section 4 includes it explicitly.

3.4 Codewords and SFD

The two-section model applies uniformly to all symbol types, differing only in the codeword c and the chirp direction v :

- **Preamble upchirps** ($c = 0, v = +1$): The code-dependent term $f_{\text{mod}} = 0$, so F_1 encodes $f_{cfo} + \beta'v\tau_i$ and the phase follows Eq. 7. All preamble windows produce identical frequency patterns (modulo the progressive drift τ_i), making preamble detection a pattern-matching problem.
- **Sync-word upchirps** ($c = c'_1, c'_2; v = +1$): The known sync-word codes shift F_1 by $f_{\text{mod}} = (c'/N) \cdot BW'$. Since c' is known *a priori*, these symbols provide additional verification constraints.
- **SFD downchirps** ($c = 0, v = -1$): The chirp sweeps in the opposite direction, so the TO contribution to F_1 reverses sign: F_1 encodes $f_{cfo} - \beta'v\tau_i$ instead of $f_{cfo} + \beta'v\tau_i$. The quadratic coefficient also flips sign. This sign reversal provides a complementary frequency-domain measurement.
- **Payload upchirps** ($c = c_i$, unknown; $v = +1$): The code c_i is the data to be decoded. The same (F_1, P_1, F_2, P_2) model applies, but now c_i is unknown and must be estimated.

3.5 Summary: What (f_{cfo}, τ_0) Determines

The signal model establishes that the pair (f_{cfo}, τ_0) jointly and deterministically controls:

- (1) **Signed frequency measurements.** From the upchirp and downchirp dechirping results $f_{\uparrow}$ and $f_{\downarrow}$, $f_{cfo} = (f_{\uparrow} + f_{\downarrow})/2$ and $\tau_0 = (f_{\uparrow} - f_{\downarrow})/(2\beta)$. These spectral values are initially sampled on the FFT grid and therefore have finite granularity. With a zero-padding factor L , the grid spacing is $BW/(LN)$, while the dechirped tone has a sinc-shaped peak with null-to-null width $2BW/N$. Consequently, at least one FFT sample falls close to the peak. The worst-case half-grid offset incurs a scalloping loss of $20 \log_{10}[\text{sinc}(1/(2L))]$, allowing the coarse FFT to retain most of the signal energy. The subsequent inter-chirp phase estimate refines the CFO to a continuous value rather than restricting it to this FFT grid.
- (2) **The two-section decomposition** (F_1, P_1, F_2, P_2) for every symbol window (Eqs. 10–11).
- (3) **The inter-symbol phase progression.** The frame phase $\phi_i = \phi_0 + 2\pi f_{cfo} \cdot i \cdot T'$ increases linearly across

preamble symbols with a slope determined by f_{cfo} (Eq. 7).

The coherent detection score is obtained by summing all sections across all symbol windows with their predicted phases reverted:

$$S(f_{cfo}, \tau_0) = \left| \sum_{\text{all sections}} V_i \cdot e^{-jP_i} \right| \quad (12)$$

where V_i is the FFT value at the predicted bin and P_i is its predicted section phase (P_1 or P_2). This score is maximized when (f_{cfo}, τ_0) matches the true values, achieving constructive coherent combination. The question is how to search efficiently—the subject of Section 4.

4 Phase-Coherent Detection and Decoding Algorithm

Section 3 showed that the pair (f_{cfo}, τ_0) determines the frequency and phase of every dechirped symbol through the two-section model (F_1, P_1, F_2, P_2) . The coherent score $S = |\sum V_i \cdot e^{-jP_i}|$ (Eq. 12) is maximized at the true (f_{cfo}, τ_0) . DeRa first dechirps each preamble chirp and computes its FFT, and then applies a column-wise FFT across the chirps to obtain coherent-power peaks and a continuous fractional-CFO estimate. It then resolves the bounded CFO aliases and timing offset, coherently verifies the result, and performs phase-compensated payload decoding with a bounded CRC-guided fallback. The per-symbol FFT, column-wise FFT, and refinement form a coarse-to-fine search for the maximum coherently accumulated power, which occurs when the correct CFO aligns the inter-chirp phases; thus, the resulting peak robustly identifies the true CFO.

4.1 Two-Stage FFT for Coherent Detection

Key observation: For the K preamble symbols used for detection ($c = 0$), the frame phase $\phi_i = \phi_0 + 2\pi f_{cfo} \cdot i \cdot T'$ increases linearly in the symbol index i (Eq. 7). This means the sequence of FFT peak values $\{Y_i[k^*]\}_{i=0}^{K-1}$ forms a complex sinusoid in i with frequency $f_{cfo} \cdot T'$. A second FFT along this axis directly resolves this rotation rate.

We first build a $K \times N$ spectral matrix Y by computing the N -point FFT of each dechirped preamble symbol. Identify the peak column k^* , which is the bin with maximum total power across all rows. We then compute a zero-padded FFT of length M along this column:

$$Z[m] = \sum_{i=0}^{K-1} Y_i[k^*] \cdot e^{-j2\pi mi/M}, \quad m = 0, \dots, M-1 \quad (13)$$

The peak m^* encodes the inter-symbol phase rotation rate, yielding f_{cfo} modulo BW/N :

$$\hat{f}'_{cfo} = \frac{m_{\text{eff}}}{M \cdot T}, \quad m_{\text{eff}} = \begin{cases} m^* & m^* \leq M/2 \\ m^* - M & m^* > M/2 \end{cases} \quad (14)$$

The column-wise FFT provides $f_{cfo} \bmod BW/N$:

$$f_{cfo}^{(n)} = \hat{f}'_{cfo} + n \cdot \frac{BW}{N}, \quad |f_{cfo}^{(n)}| \leq f_{cfo, \max} \quad (15)$$

To recover the exact CFO value, we compute padded (P -fold zero-padding) per-symbol FFTs and extract DC-centered signed frequency measurements (see Appendix B):

$$f_{\uparrow} = f_{cfo} + \beta\tau_0 \in [-BW/2, BW/2] \quad (16)$$

$$f_{\downarrow} = f_{cfo} - \beta\tau_0 \in [-BW/2, BW/2] \quad (17)$$

The coarse CFO estimate $\hat{f}_{cfo}^{\text{coarse}} = (f_{\uparrow} + f_{\downarrow})/2$ is accurate to $BW/(NP)$ but unaliased. We fuse it with the fine estimate to obtain the optimal $\hat{f}_{cfo}$ candidate:

$$\hat{f}_{cfo} = \hat{f}'_{cfo} + \text{round}\left(\frac{\hat{f}_{cfo}^{\text{coarse}} - \hat{f}'_{cfo}}{BW/N}\right) \cdot \frac{BW}{N} \quad (18)$$

The time offset is then: $\hat{\tau}_0 = \frac{f_{\uparrow} - \hat{f}_{cfo}}{\beta}$

4.2 Coherent CFO/TO Refinement

We retain the 20 best FFT-derived $(\hat{f}_{cfo}, \hat{\tau}_0)$ candidates and compute coherent scores for them. We first evaluate (F_1, P_1, F_2, P_2) expressions (Eqs. 10–11), and use them to gather FFT values from the precomputed spectra (one high-resolution FFT per symbol window, computed once). The coherent sum is accumulated in complex double precision:

$$S[c, m] = \sum_{\text{all sections}} V_{c,m} \cdot e^{-j(P_{c,m} + 2\pi\delta f_{c,m} \cdot t_{\text{mid}})} \quad (19)$$

The detection score is $|S[c, m]|/N_s$.

We refine each of the 20 candidates over a local neighborhood of five CFO and five timing offsets, interpolate its power maximum, and rerank by refined power.

Computational complexity. The precomputation step computes $(K + 5)$ FFTs (one per symbol window) of length N_{fft} , costing $O(K \cdot N_{\text{fft}} \log N_{\text{fft}})$. The column-wise FFT costs $O(NM \log M)$, where $M \ll N$. The refinement phase performs a gather-and-sum operation that phase-aligns K values from the precomputed FFT results. Evaluating 20 candidates over five CFO and five timing offsets requires $20 \times 5 \times 5 = 500$ such operations. Its fixed search size does not change the overall $O(KN \log N)$ complexity, or $O(N \log N)$ per symbol. These hyperparameters can be tuned to achieve a tradeoff between computational cost and noise robustness. Overall, the reusable FFT precomputation, column-wise FFT, and bounded refinement have complexity $O(KN \log N)$, or

$O(N \log N)$ per symbol. Thus, DeRa has the same asymptotic complexity as standard FFT-based detection, although it incurs higher constant-factor computation. A complete pseudocode summary is provided in Appendix C.

4.3 Phase-Coherent Payload Decoding

With the estimated $(\hat{f}_{cfo}, \hat{\tau}_0)$ from detection, we decode the payload in multiple stages of increasing sophistication.

Stage 1: Phase-compensated magnitude decode. We dechirp each payload symbol with standard downchirps and compute the FFT. Because a payload symbol carries code c , the chirp wraps around at fraction $1 - c/N$ within the symbol, creating a *phase discontinuity* that splits the energy across two FFT bins. We compute this phase jump *analytically* from the preamble-derived $(\hat{f}_{cfo}, \hat{\tau}_0)$. It contains two parts:

- (1) **CFO-induced phase accumulation.** The first chirp segment spans a duration $T'(1 - c/N)$. Over this segment, the CFO accumulates a phase offset:

$$\Delta\phi_{cfo}[c] = 2\pi f_{cfo} \cdot T' \cdot (1 - c/N) \quad (20)$$

- (2) **Time-offset-induced frequency shift.** The accumulated time offset shifts the wrap boundary by $\beta(\tau_0 - i\epsilon T)$ in frequency. This produces a phase that depends on both the code position and the symbol index within the payload:

$$\Delta\phi_{TO}[c] = 2\pi \cdot BW \cdot (T' - T) \cdot \left(1 - \frac{c}{N} + N_{pre} + 4.25 + k\right) \quad (21)$$

where k is the index of the payload symbol (zero-based).

Both terms are determined by the known parameters $(\hat{f}_{cfo}, \hat{\tau}_0, k)$ and the to-be-searched parameter c .

Coherent recombination. Denoting the two FFT contributions as V_1 (first segment) and V_2 (second segment), we compensate for the phase jump:

$$Y_i^{\text{coh}}[c] = V_1[c] + V_2[c] \cdot e^{-j(\Delta\phi_{cfo}[c] + \Delta\phi_{TO}[c])} \quad (22)$$

This coherently adds the two segments at full energy, in contrast to LoRaTrimmer's incoherent $|V_1|^2 + |V_2|^2$ which loses up to 3 dB. The decoded symbol is the magnitude peak of the coherently combined spectrum:

$$\hat{c}_i = \arg \max_c |Y_i^{\text{coh}}[c]| \quad (23)$$

Decode the result and verify it via LoRa's built-in CRC. If CRC passes, accept the decode.

Stage 2: Phase-coherent decode (if CRC fails). The magnitude-based method discards phase information. When the initial decode fails, we exploit the phase model for improved decoding. We estimate the most probable initial phase $\hat{\phi}_0$ of the payload from the complex sum of all selected and phase-compensated FFT peaks: $\hat{\phi}_0 = \angle(\sum_i Y_i[\hat{c}_i])$, and predict the per-symbol phase: $\hat{\phi}_i = \hat{\phi}_0 + 2\pi \hat{f}_{cfo} (N_{pre} + 4.25 + i) \cdot T'$

using Eq. 7. We re-decode via real-part projection: $\hat{c}_i = \arg \max_k \text{Re}(Y_i[k] \cdot e^{-j\hat{\phi}_i})$.

Stage 3: Alternative CFO candidates (if CRC still fails). If the phase-coherent decode also fails, the most likely cause is an incorrect CFO estimate. We fall back to the next-ranked candidate from the detection stage and repeat Stages 1–2 with the updated $(\hat{f}_{cfo}, \hat{\tau}_0)$. We try at most five candidates.

4.4 Unified Detection–Decoding Pipeline

The complete pipeline embodies a key insight: *detection and decoding are two aspects of a single, CFO-driven process*. The CFO estimated during detection is the same physical quantity that determines payload phase distortions; the CRC outcome during decoding feeds back to validate or reject that estimate.

- (1) **Detect:** Column-wise FFT + coarse–fine fusion $\rightarrow$ ranked (f_{cfo}, τ_0) candidates.
- (2) **Decode:** For each candidate in rank order:
 - (a) Magnitude decode (Eq. 23) $\rightarrow$ CRC check.
 - (b) If fail: phase-coherent decode $\rightarrow$ CRC check.
 - (c) If fail: next candidate.
- (3) **Accept** the first CRC-passing result, or report detection failure after exhausting all candidates.

By treating the entire packet as one coherent entity connected by CFO, we simultaneously achieve more sensitive detection and more accurate decoding than any pipeline that treats them in isolation.

5 Evaluation

5.1 Experimental Setup

Hardware. We use two COTS LoRa transmitters—Semtech SX1276 [42] and SX1280 [43]—and receive with a USRP N210 [16] SDR equipped with SBX 400–4400 MHz RF daughterboards [15], controlled via UHD and GNU Radio at a sampling rate F_s of 1 MHz, a representative sampling frequency for COTS gateways [44, 45]. We evaluate two configurations: (i) 903.9 MHz carrier with 125 kHz bandwidth (SX1276) and (ii) 2.4 GHz carrier with 203.125 kHz bandwidth (SX1280), both with $\text{SF} \in \{7, 8, 9, 10, 11, 12\}$, 8-symbol preambles, code rate $\text{CR} = 4/8$, and approximately 16-byte payloads transmitted periodically. All DeRa processing runs on a Raspberry Pi 4 equipped with four 1.5-GHz Cortex-A72 cores.

Deployment. Experiments are conducted across indoor and outdoor environments, as shown in Figure 4. The indoor setting is an office with furniture, computers, and regular human activity that induces dynamic channel variations. We use two outdoor environments: (i) a university campus with buildings, pedestrians, and vehicular traffic, and (ii) a farm with fewer obstructions but scattered trees. We add

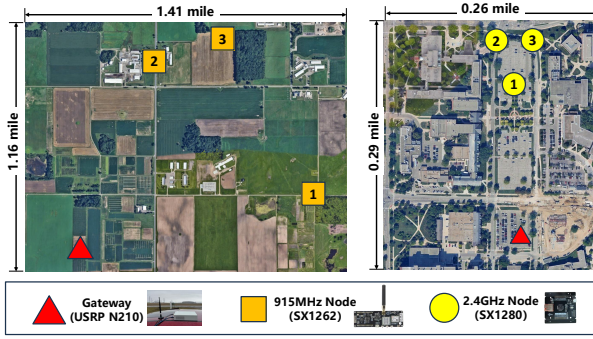


Figure 4: Deployments in farm and campus.

calibrated synthetic noise to captured signals to evaluate performance across specific SNR levels.

Baselines. We compare against two state-of-the-art end-to-end detection-and-decoding systems: CloudLoRa [46], a non-coherent energy-based detector (abbreviated **CL** in figures) and LoRaTrimmer [12], who partially mitigate phase misalignment with a probabilistic model (abbreviated **LT** in figures). We re-implement both baselines and run all three methods on the identical captured signals to ensure a fair comparison. DeRa is opensourced at <https://github.com/daibiaoxuwu/DeRa>.

Metrics. We report three metrics: Packet Detection Rate (PDR), Symbol Error Rate (SER), and Packet Reception Rate (PRR). PDR counts a detection as correct only when the CFO estimation error is within $BW/2^{SF}$ and timing offset error within $1/BW$ (which are the tolerances required for successful decoding). SER is the raw demodulation error rate before Hamming correction. PRR is the end-to-end success rate after Hamming correction and CRC verification. We use the *90% threshold SNR* (minimum SNR for 90% PDR or PRR) and *10% threshold SNR* (minimum SNR for $SER < 10\%$) to quantify SNR gains between methods.

5.2 Detection Performance

Figure 5(a) shows that DeRa outperforms LoRaTrimmer and CloudLoRa in all metrics at all SNRs. Furthermore, as SNR decreases, the PDR of DeRa degrades more gradually than LoRaTrimmer, which in turn drops more slowly than CloudLoRa. This ordering directly reflects preamble accumulation coherence: DeRa adds symbol energies phase-coherently, so the accumulated power grows quadratically with the number of symbols; LoRaTrimmer’s probabilistic model achieves only partially coherent gain; CloudLoRa’s magnitude-only addition yields the weakest accumulation. Figure 5(b) quantifies this gap: DeRa lowers the 90% PDR threshold SNR by 4.3–4.8 dB over LoRaTrimmer and 5.3–6.5 dB over CloudLoRa across all SFs.

5.3 Decoding Performance

The same coherent-processing advantage extends to decoding. Because DeRa performs phase-coherent demodulation—constructively combining signal energy across chirp symbols—it achieves lower SER and higher PRR than the partially coherent model of LoRaTrimmer and the non-coherent demodulation of CloudLoRa. As shown in Figure 5(c,d) and Figure 6(a,b), DeRa improves the 10% SER threshold by 3.3–3.9 dB over LoRaTrimmer and 4.4–5.2 dB over CloudLoRa, and lowers the 90% PRR threshold by 2.9–3.7 dB and 4.8–5.7 dB, respectively, with consistent gains across every SF. To isolate the decoder contribution from detection, we further evaluate all three decoders under perfect CFO and TO knowledge (Figure 6(c,d)). Even with identical synchronization inputs, DeRa lowers the 10% SER threshold by 1.9–2.0 dB over LoRaTrimmer and 4.6–4.7 dB over CloudLoRa, and lowers the 90% PRR threshold by 1.7–1.9 dB and ~ 3.1 dB, respectively, confirming that a substantial portion of the end-to-end gain originates from the accurate estimation of CFO and TO and the phase-coherent decoder.

5.4 CFO Robustness

We evaluate detection performance across five CFO levels: 0, 500, 1,000, 10,000, and 40,000 Hz. The two endpoints are from real hardware—the SX1276’s TCXO yields near-zero CFO, while the SX1280’s 20 ppm oscillator produces up to 40 kHz at 2.4 GHz—and the three intermediate values are synthetically injected to fill the gap. We test at two near-threshold settings: SF7/−17 dB and SF10/−24 dB (Figure 7(a,b)).

DeRa maintains the highest and most stable PDR and PRR across the entire CFO range because it jointly estimates and compensates CFO-induced distortions before accumulation. CloudLoRa performs well near zero CFO because its detection algorithm implicitly assumes no frequency offset; however, this same assumption causes it to collapse rapidly as CFO increases, since its non-coherent accumulation cannot compensate for the resulting frequency misalignment and phase shifts. LoRaTrimmer’s probabilistic model partially mitigates phase shifts, keeping its PDR relatively stable at SF7 (see Figure 7(a)). However, at SF10 its PRR drops sharply at large CFO (see Figure 7(b)): the denser frequency bins ($BW/2^{SF}$) at higher SFs leave less tolerance for residual CFO estimation error, and the continuous frequency shift caused by CFO accumulates across payload symbols until it exceeds one bin width, causing massive decoding failures. In contrast, DeRa mitigates CFO-induced distortions, maintaining stable PDR and PRR at all CFO levels.

Figure 7(c) sweeps CFO across one FFT-bin interval at SF7/−18 dB. The periodic ripple reflects the combined resolution of the two-stage FFT; PDR and PRR deviate from

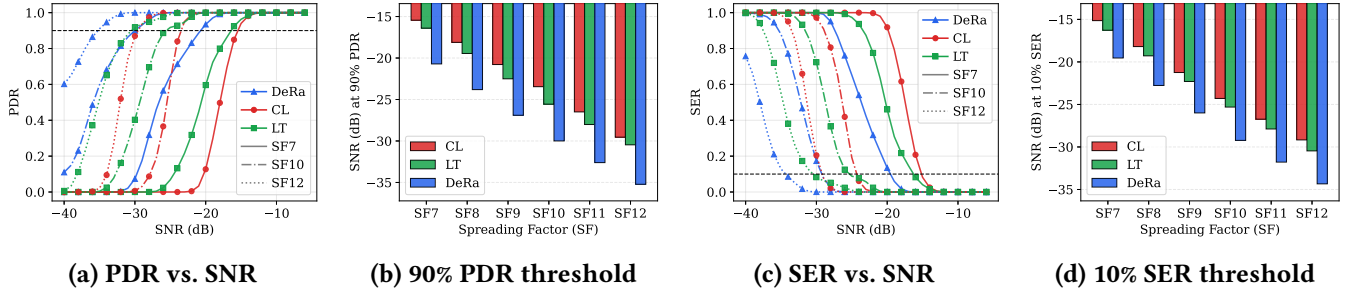


Figure 5: Detection and decoding performance: PDR and SER. Curves show $SF \in \{7, 10, 12\}$; bars show threshold SNR across all SFs (lower is better).

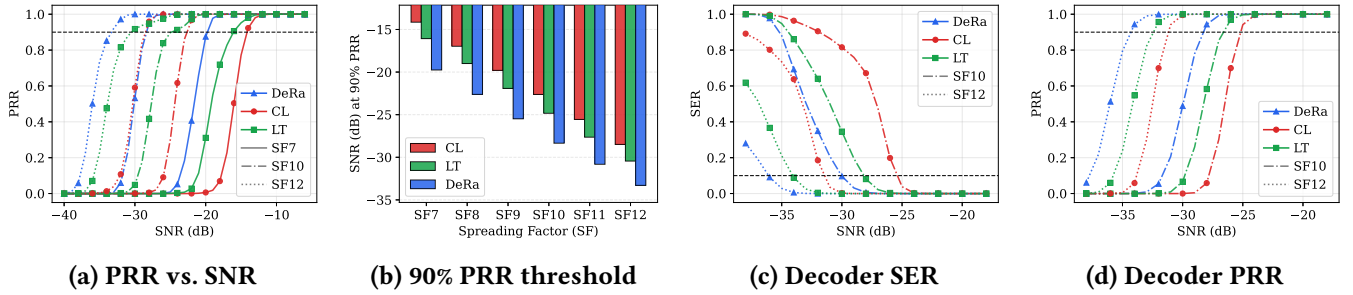


Figure 6: Detection and decoding performance: PRR and decoder evaluation. (a,b) End-to-end PRR; (c,d) decoder performance under perfect CFO and TO knowledge.

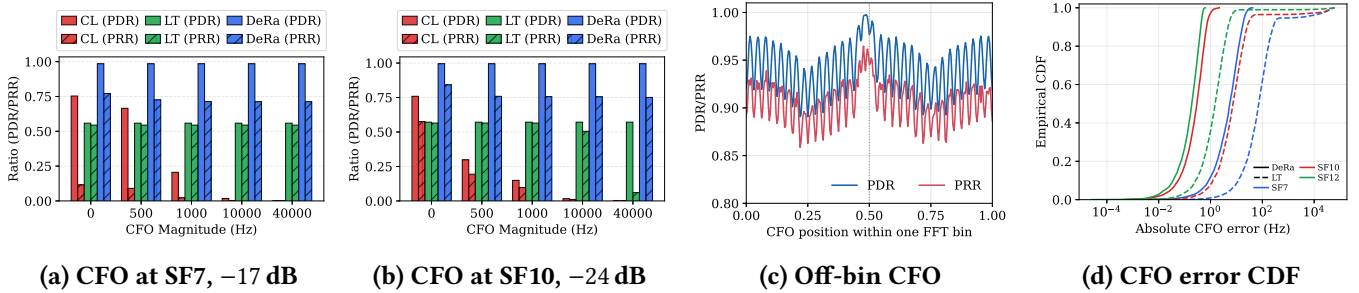


Figure 7: CFO robustness and estimation. (a,b) PDR and PRR at different CFO magnitudes. (c) PDR and PRR at fractional CFO positions. (d) Absolute CFO estimation errors for DeRa and LoRaTrimmer.

the center by at most 5.35% and 5.30%, respectively, confirming that off-bin CFO has little effect on synchronization. Figure 7(d) compares the absolute CFO-error CDFs of DeRa and LoRaTrimmer at SF7/-16 dB, SF10/-24 dB, and SF12/-28 dB. DeRa’s 90th-percentile error is 16.99, 0.70, and 0.42 Hz, respectively, remaining below the corresponding FFT-bin spacing (488.28, 61.04, and 30.52 Hz) and outperforming the LoRaTrimmer baseline.

5.5 Ablation Study

We first isolate the detection components in Figure 8(a). Replacing phase-coherent accumulation with magnitude-only addition (“Abs-add”) reduces PDR, and additionally removing CFO-induced scaling-effect mitigation (“noCFO”) causes

a further reduction. These results show that both phase-coherent accumulation and CFO-distortion modeling contribute to detection sensitivity. The “Sub-1 only” variant retains only the first section of each split receiver window. Its PDR and PRR remain below 82% and 83%, respectively, confirming that the discarded section contains signal energy needed for coherent accumulation.

For decoding, Figure 8(b) shows that disabling CFO fine-tuning based on preamble-phase regression (“noFinetune”) lowers PRR. Replacing the remaining decoder with LoRaTrimmer’s demodulator widens the gap, demonstrating the additional contribution of DeRa’s phase-coherent decoder.

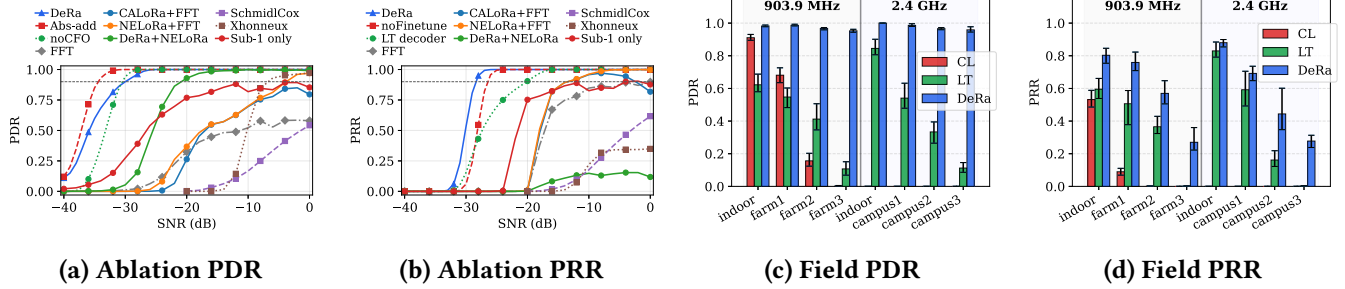


Figure 8: Component ablations, alternative receivers, and field performance. (a) PDR performance. (b) PRR performance. (c,d) Field PDR and PRR at the indoor, farm, and campus sites.

We also compare alternative receiver pipelines. In Figure 8(a), CALoRa+FFT uses CALoRa’s self-attention detector [26], whereas NLoRa+FFT applies NLoRa’s MaskCNN denoiser [28] before FFT-based detection. Both learned pipelines outperform the standard FFT receiver, but require substantially higher SNR than DeRa. Large CFO and unknown window alignment make it difficult for neural models to learn robust discriminative features. The Schmid–Cox method [41] estimates synchronization using the correlation between repeated signal segments, while Xhonneux et al. [60] estimate CFO and timing offsets from consecutive LoRa preamble chirps. Both degrade at low SNR because these synchronization features are sensitive to noise. In Figure 8(b), DeRa+NLoRa connects the detector of DeRa to NLoRa’s decoder. Its improvement over the NLoRa+FFT baseline shows the value of DeRa’s detection stage, while its lower PRR than the full DeRa pipeline isolates the benefit of DeRa’s phase-coherent decoder.

5.6 Real-World Field Experiments

We deploy both COTS transmitters—SX1276 at 903.9 MHz (TCXO, near-zero CFO) and SX1280 at 2.4 GHz (20 ppm oscillator, up to 40 kHz CFO)—at varying indoor and outdoor distances using SF 10. As shown in Figure 8(c), DeRa sustains 95.5%–100% PDR across all locations regardless of CFO magnitude. CloudLoRa drops to near zero at outdoor sites where the SX1280’s large CFO overwhelms its no-CFO assumption, and LoRaTrimmer falls below 11% at the farthest locations where low SNR compounds residual CFO errors. The same pattern appears in PRR (Figure 8(d)): DeRa delivers the highest end-to-end reception at every site, confirming that its gains translate from controlled experiments to real-world deployments.

5.7 Effects of Preamble Lengths

Figure 9(a) shows that increasing the preamble length lowers the SNR required to reach 90% PDR, with diminishing returns because each doubling provides at most 3 dB of additional coherent energy. DeRa supports variable preamble

lengths, including LoRa’s six-symbol minimum [42]. Across SF7–SF12 and preamble lengths of 6–16 symbols, the per-window runtime utilization ranges from 45.8% to 60.9% on a single Raspberry Pi CPU core, demonstrating that DeRa operates in real time across the evaluated preamble lengths.

5.8 Robustness to Time-Varying CFO, Phase Noise, and Interference

Figure 9(b) compares the PRR robustness of DeRa and LoRaTrimmer under within-packet frequency variation. We simulate oscillator-frequency drift as a random walk whose per-sample frequency increments have standard deviation $D_f/\sqrt{F_s}$. We additionally apply Wiener phase noise with per-sample increment standard deviation σ , fixed such that $\sigma\sqrt{F_s} = 1 \text{ rad}/\sqrt{s}$. We evaluate SF7 at -16 dB , SF10 at -24 dB , and SF12 at -28 dB . Higher spreading factors produce longer packets, allowing more frequency drift to accumulate and thereby degrading performance. From $D_f = 0$ to $8 \text{ Hz}/\sqrt{s}$, the PRR of DeRa decreases by 0.1, 0.5, and 14.8 percentage points at SF7, SF10, and SF12, and is substantially more robust than LoRaTrimmer. We further evaluate robustness under Doppler effects and mobility by mounting the device on a car and driving near the fixed campus and farm positions. Mobility reduces PDR/PRR by 4.4%/5.2% at SF7 and 6.8%/7.9% at SF10, indicating that DeRa remains robust to practical frequency variation and phase noise.

Figure 9(c,d) reports the corresponding sweeps with asynchronous four-node traffic at SIRs of -3 , -6 , and -9 dB . PRR decreases as SIR falls because interfering symbols affect payload decoding, whereas PDR remains comparatively stable, demonstrating DeRa’s detection robustness under interference.

5.9 Computation–Performance Tradeoff

Figure 10(a,b) compares the full PDR and PRR curves for SF7, SF10, and SF12 under three computation budgets, implemented by scaling the sizes of the two-stage FFT to 0.25 $\times$, 1 $\times$, and 4 $\times$ their default values. The 4 $\times$ setting improves the

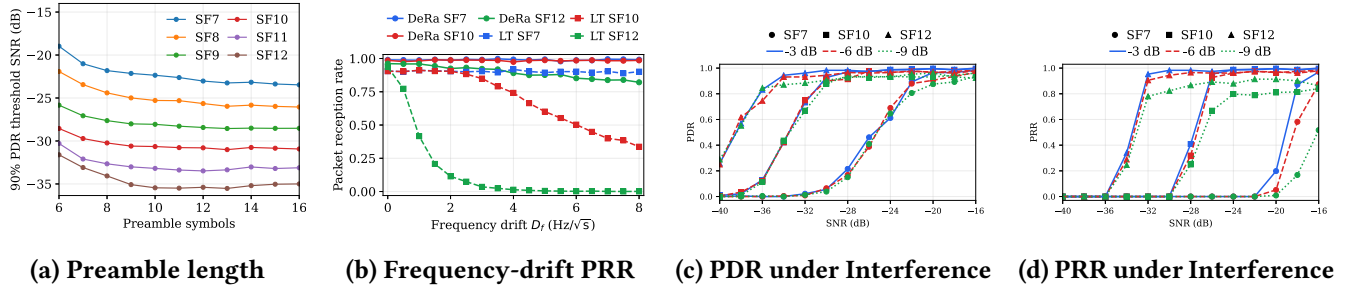


Figure 9: Adaptiveness to different preamble length, and robustness against frequency drift and interference. (a) SF7–SF12 90%-PDR thresholds versus preamble length. (b) DeRa and LoRaTrimmer PRR under within-packet frequency drift. (c,d) PDR and PRR with asynchronous four-node traffic at SIRs of -3 , -6 , and -9 dB.

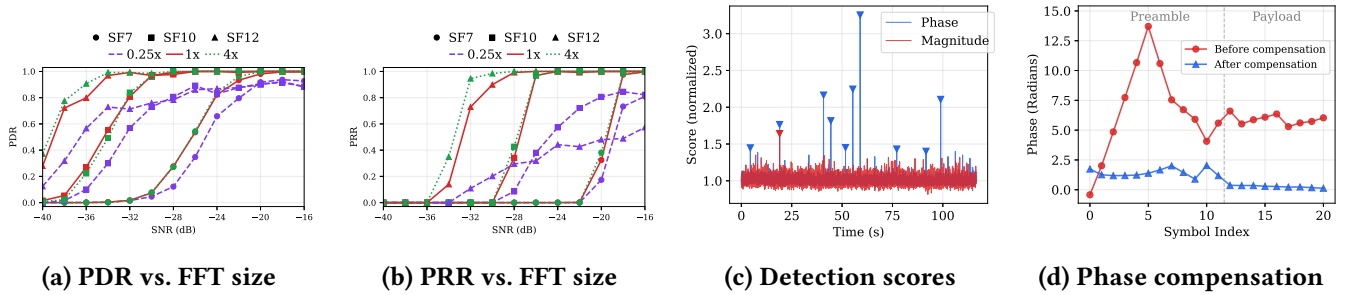


Figure 10: Tradeoff between computation and performance, and phase-coherent detection case studies. (a,b) PDR and PRR at $0.25\times$, $1\times$, and $4\times$ computation budgets. (c) Magnitude-only and phase-coherent detection scores in a field deployment. (d) Per-symbol phase before and after CFO compensation.

average 90%-PDR and 90%-PRR thresholds by only 0.62 dB and 0.75 dB, respectively, while increasing runtime by 269% on average. The $0.25\times$ setting reduces runtime by 24% but substantially degrades both PDR and PRR. Thus, the default $1\times$ setting provides the best tradeoff between computation and reception performance.

5.10 Case Study on Phase-Coherent Detection

We illustrate the benefits of phase-coherent detection with an SX1280 node transmitting periodically at 2.4 GHz, SF12, at 0.25 miles on campus under heavy occlusion. Figure 10(c) compares the normalized scores of non-coherent magnitude-only detection (“Magnitude”) and phase-coherent detection (“Phase”). Phase-coherent accumulation produces higher, more distinct peaks by concentrating signal energy instead of spreading it across the noise floor. Figure 10(d) shows the unwrapped per-symbol phase of one detected packet. CFO causes a linear phase increase before compensation; afterward, the phase remains approximately constant across the preamble and payload, enabling coherent detection and decoding.

5.11 Runtime and Computational Cost

Table 1 reports detector runtime per window for DeRa and the baselines on a Raspberry Pi 4B using one CPU core. Real-time operation requires the computation time to remain below the symbol duration it processes. DeRa uses 46.5–55.7% of this time budget across SF7–SF12, slower than the standard FFT method but faster than LoRaTrimmer and Cloud-LoRa. The Dechirp/FFT, column-wise FFT, and coherent candidate scoring account for 9.6%, 37.2%, and 53.2% of runtime on average, respectively. When simultaneously receiving multiple channels, two/four/six/eight channels consume 23.57/48.86/74.15/98.66% of the four-core runtime budget; all evaluated channel counts operate in real time. Under controlled CPU, system-call, and synchronous-I/O contention, DeRa has a 99th-percentile latency of 91.5% of the symbol duration across the tested load profiles.

6 Related Work

Weak Signal Detection. Detecting LoRa packets below the noise floor is the primary bottleneck for extending communication range [1, 17, 22, 35, 54, 56, 60, 68]. Recent works improve weak signal reception via chirp trimming [12], parabolic phase modeling [28], spectrogram features [40], or extending preamble structures [29, 70]. Neural networks

Table 1: Mean per-symbol detector service time on a Raspberry Pi 4B using one CPU core (ms). The three indented rows apportion DeRa’s fused runtime using separately measured component shares.

Method	SF7	SF8	SF9	SF10	SF11	SF12
Symbol duration	1.024	2.048	4.096	8.192	16.384	32.768
DeRa	0.476	0.960	1.972	4.525	9.034	18.260
Dechirp/FFT	0.079	0.115	0.168	0.312	0.508	1.449
Column-wise FFT	0.109	0.263	0.640	1.836	4.568	8.964
Coherent scoring	0.288	0.582	1.164	2.377	3.958	7.847
Standard FFT	0.096	0.133	0.213	0.385	0.812	2.274
LoRaTrimmer	6.859	12.172	23.727	54.346	120.624	269.377
CloudLoRa	49.955	703.709	2,435.02	8,662.50	30,309.8	150,840

have also been applied to LoRa signal detection [48]. Tesfay et al. [50] combine autoencoder-based denoising with CNN-based symbol detection in multiuser LoRa-like uplinks. More recently, CALoRa [26] uses a self-attention-based chirp restorer to reconstruct noisy I/Q spectrograms, followed by a convolutional detector that localizes the preamble.

Coherent Reception and Synchronization. Afisiadis et al. [1] analyze coherent LoRa symbol detection when phase information is available. Xhonneux et al. [60] estimate fractional CFO and STO from consecutive preamble upchirps. They target coherent demodulation and synchronization rather than weak-packet acquisition. The Schmidl–Cox OFDM method [41] jointly detects a specially designed repeated training sequence and estimates timing and CFO. In contrast, DeRa jointly detects the packet and estimates CFO through coherent accumulation at deep-negative SNR.

CFO and RF Impairments. CFO is traditionally treated as a hardware impairment to be exploited for device fingerprinting and collision resolution [14, 58] or corrected after packet detection to prevent symbol errors [49, 68]. Instead of merely tracking CFO as a fingerprint or nuisance, DeRa transforms CFO into an asset: leveraging the deterministic phase evolution it imposes to achieve phase-coherent accumulation and boost detection sensitivity. LoRa radios also expose frequency-error indicators for Automatic Frequency Correction (AFC) after usable reception [42]. AFC can estimate frequency error and retune subsequent frames, whereas DeRa focuses on coherent detection of the current weak packet; the two mechanisms are complementary.

Sensitivity via Spatial and Resource Diversity. Several designs increase detection sensitivity by combining multiple observations across distributed gateways [2, 10, 13, 31, 34, 46], collocated gateway antennas [22], or retransmitted packets [61]. In contrast, DeRa unlocks the coherent processing gain already latent in every LoRa preamble, achieving 4.3–4.8 dB gains in detection sensitivity.

Collision, Concurrency, and Robustness. Beyond pushing sensitivity, packet detection is also addressed in collision recovery and massive concurrency [6, 7, 14, 47, 51, 52, 57, 59,

62, 64, 65, 69, 73]. These systems assume reliable boundary detection and could potentially benefit from DeRa’s enhanced front-end. Other efforts focus on robust packet detection under diverse challenges, including interference and spectrum dynamics [5, 25, 50, 55, 66, 67], hardware constraints [63], scalable gateway architectures [24, 71, 72], jamming [21], and backscatter scenarios [19, 23, 38]. While these address specific environmental or scale constraints, DeRa targets the fundamental, physical-layer sensitivity limit of ultra-weak packet detection.

7 Discussion

DeRa Integration in Practice: DeRa indeed trades a higher computational cost than a standard FFT-based receiver for improved detection and decoding at low SNR. This additional cost arises primarily from the column-wise FFT and phase-alignment operations. Our experimental results show that the resulting receiver can still operate in real time on a Raspberry Pi or comparable hardware, suggesting that DeRa can be deployed on low-cost platforms akin to commercial LoRa gateways with sophisticated software-hardware co-design to further optimize computation efficiency and simultaneously support multiple data streams from different channels.

When more demanding workloads, such as simultaneously detecting more than eight channels, are required, similar to CloudLoRa [46], the main computation tasks of DeRa can be offloaded to an edge or cloud server. Specifically, local LoRa gateways use the standard FFT-based receiver to handle detectable signals and offload signals that failed detection or decoding to a DeRa server. Although this approach alleviates the local computation burden, it consumes considerable network bandwidth on the backhaul between the gateways and the DeRa server, posing a challenge for designing the offloading strategy.

8 Conclusion

We present DeRa, a phase-coherent detection and decoding framework that jointly estimates CFO and compensates CFO-induced phase distortions, enabling both sensitive packet detection and phase-aligned payload decoding. Experiments show that DeRa lowers the detection threshold by 4.3–4.8 dB and the decoding threshold by 3.3–3.9 dB over state-of-the-art baselines, while remaining robust under CFOs up to 40 kHz. These results demonstrate that phase-coherent processing can substantially improve LoRa sensitivity and robustness without requiring more expensive hardware.

9 Acknowledgments

We thank the reviewers and our shepherd for their guidance. This work was supported in part by the National Science Foundation under Grant Nos. 2338976 and 2312674.

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A Signal Model: Detailed Derivations

This appendix provides the complete mathematical derivations underlying the signal model of Section 3. We expand on the global phase-continuity framework, the accurate frequency characteristics for each symbol in the LoRa packet, the per-symbol phase increment proof, the derivation of the dechirped residual phase, and the linear-phase approximation that enables efficient FFT-based detection.

A.1 Global Phase-Continuity Framework

The transmitted LoRa signal is a piecewise sequence of M segments (full or partial symbols), where the j -th segment occupies the time interval $[t_j, t_{j+1}]$ with instantaneous frequency $f_j(t)$. The instantaneous phase at absolute time t within this segment is:

$$\Phi(t) = 2\pi \int_{t_j}^t f_j(\tau) d\tau + \phi_j, \quad t \in [t_j, t_{j+1}] \quad (24)$$

Phase continuity across segment boundaries requires $\phi_{j+1} = \Phi(t_{j+1})$. This constraint is central: it threads a single, unbroken phase trajectory through the entire LoRa frame, linking every symbol—preamble, sync word, SFD, and payload—into one coherent signal model.

LoRa frame structure. Under the shared-clock model, each full symbol has exact receiver-time duration $T' = T/(1+\epsilon)$. The implementation uses $T(1-\epsilon)$ as its first-order update. The complete frame comprises:

- (1) **Preamble:** N_{pre} upchirps with $c = 0$. Total duration: $N_{pre}T'$.
- (2) **Sync word:** 2 coded upchirps with $c = c'_1, c'_2$. Total duration: $2T'$.
- (3) **SFD:** 2.25 downchirps. Total duration: $2.25T'$.
- (4) **Payload:** N_{pay} coded upchirps with $c = c_1, \dots, c_{N_{pay}}$. Total duration: $N_{pay}T'$.

The phase-continuity constraint ensures that the phase at the boundary of any two adjacent segments is unbroken, regardless of whether the transition is between symbols of the same type (e.g., consecutive preamble chirps), between different types (e.g., preamble to sync word), or across the frequency wrap point within a single chirp.

A.2 Piecewise Frequency Definitions

We define the instantaneous frequency $f_j(t)$ for each symbol type under CFO.

Upchirps (preamble, sync word, and payload). A LoRa upchirp with integer cyclic shift $c \in \{0, \dots, N-1\}$ has normalized shift $\tilde{c} = c/N \in [0, 1)$, with $c = 0$ for preamble base chirps. It sweeps upward at rate β' and folds (wraps by $-BW'$) at $t^* = T_{start} + (1-\tilde{c})T'$ to stay within the bandwidth.

For the symbol starting at $T_{start} = iT'$:

$$f_i(t) = \begin{cases} BW'(\tilde{c} - 0.5) + f_{cfo} + \beta'(t - T_{start}) & t \in [T_{start}, t^*] \\ BW'(\tilde{c} - 1.5) + f_{cfo} + \beta'(t - T_{start}) & t \in [t^*, T_{start} + T'] \end{cases} \quad (25)$$

The frequency is discontinuous at t^* (jumping by $-BW'$), but the *phase* remains continuous because the integral in Eq. 24 is unaffected by instantaneous frequency jumps. To verify this, note that the phase at t^* accumulated from both pieces of the piecewise function is:

$$\begin{aligned} \Phi(t^*) &= 2\pi \int_{T_{start}}^{t^*} [BW'(\tilde{c} - 0.5) + f_{cfo} + \beta'(\tau - T_{start})] d\tau + \phi_{start} \\ &= 2\pi \left[BW'(\tilde{c} - 0.5)(1 - \tilde{c})T' + f_{cfo}(1 - \tilde{c})T' + \frac{\beta'}{2}(1 - \tilde{c})^2T'^2 \right] + \phi_{start} \end{aligned} \quad (26)$$

and continuing from t^* with the second piece (frequency reduced by BW') yields the same total integral at $T_{start} + T'$ as would be obtained from a continuous (unwrapped) sweep, because $(BW')(T') = 2^{SF}$ contributes an integer multiple of 2π .

Downchirps (SFD). For downchirp symbols, the frequency sweeps downward at rate β' :

$$f_i(t) = \frac{BW'}{2} + f_{cfo} - \beta'(t - T_{start}) \quad (27)$$

The downchirp sweeps from $+BW'/2 + f_{cfo}$ to $-BW'/2 + f_{cfo}$ over one symbol period T' . Unlike upchirps, downchirps with $c = 0$ do not require a frequency wrap within the symbol.

A.3 Per-Symbol Phase Increment: Full Derivation

We prove that *the phase increment per symbol is the same for all symbol types and all cyclic shifts*.

Upchirp case. For an upchirp with normalized cyclic shift $\tilde{c} = c/N$, the total phase accumulated over one symbol duration T' is:

$$\begin{aligned} \Delta\phi_{up} &= 2\pi \int_0^{(1-\tilde{c})T'} (BW'(\tilde{c} - 0.5) + f_{cfo} + \beta'\tau) d\tau \\ &\quad + 2\pi \int_{(1-\tilde{c})T'}^{T'} (BW'(\tilde{c} - 1.5) + f_{cfo} + \beta'\tau) d\tau \end{aligned} \quad (28)$$

Evaluating the first integral:

$$I_1 = 2\pi \left[BW'(\tilde{c} - 0.5)(1 - \tilde{c})T' + f_{cfo}(1 - \tilde{c})T' + \frac{\beta'}{2}(1 - \tilde{c})^2T'^2 \right] \quad (29)$$

Evaluating the second integral:

$$I_2 = 2\pi \left[BW'(\tilde{c} - 1.5) \cdot \tilde{c}T' + f_{cfo} \cdot \tilde{c}T' + \frac{\beta'}{2} (T'^2 - (1 - \tilde{c})^2 T'^2) \right] \quad (30)$$

Summing $I_1 + I_2$ and collecting terms:

$$\Delta\phi_{\text{up}} = 2\pi \left[BW'T'(\tilde{c}(\tilde{c} - 0.5) - \tilde{c}(1 - \tilde{c}) + (\tilde{c} - 0.5)(1 - \tilde{c}) - \tilde{c} \cdot 0.5) + f_{cfo}T' + \frac{\beta'T'^2}{2} \right] \quad (31)$$

Using the identity $BW' \cdot T' = 2^{SF}$ (an integer) and $\beta'T'^2/2 = BW'T'/2 = 2^{SF}/2$, we can simplify. The key observation is that all terms involving $BW'T'$ evaluate to integer multiples:

$$\begin{aligned} \text{BW terms} &= BW'T'[(\tilde{c} - 0.5)(1 - \tilde{c}) + \tilde{c}(\tilde{c} - 1.5)] + \frac{\beta'T'^2}{2} \\ &= 2^{SF}[\tilde{c} - \tilde{c}^2 - 0.5 + 0.5\tilde{c} + \tilde{c}^2 - 1.5\tilde{c}] + \frac{2^{SF}}{2} \\ &= 2^{SF}[-0.5] + \frac{2^{SF}}{2} = 0 \end{aligned} \quad (32)$$

Therefore all BW' -dependent terms contribute integer multiples of 2π and vanish modulo 2π :

$$\Delta\phi_{\text{up}} = 2\pi f_{cfo}T' \pmod{2\pi} \quad (33)$$

Downchirp case. For a downchirp, the total phase over T' is:

$$\begin{aligned} \Delta\phi_{\text{down}} &= 2\pi \int_0^{T'} \left(\frac{BW'}{2} + f_{cfo} - \beta'\tau \right) d\tau \\ &= 2\pi \left[\frac{BW'}{2}T' + f_{cfo}T' - \frac{\beta'}{2}T'^2 \right] \\ &= 2\pi \left[\frac{2^{SF}}{2} + f_{cfo}T' - \frac{2^{SF}}{2} \right] \\ &= 2\pi f_{cfo}T' \end{aligned} \quad (34)$$

Both upchirps and downchirps accumulate the identical phase increment $2\pi f_{cfo}T'$ per symbol, regardless of cyclic shift c or chirp direction.

Frame phase consequence. Since every symbol accumulates the same phase increment, and phase is continuous across symbol boundaries, the phase at the start of the j -th symbol in the frame is:

$$\phi_j = \phi_0 + 2\pi f_{cfo} \cdot j \cdot T' \quad (35)$$

For example, the initial phase of the first payload symbol ($j = N_{pre} + 2 + 2.25$) is $\phi_{\text{payload}} = \phi_0 + 2\pi f_{cfo}(N_{pre} + 4.25)T'$.

A.4 Dechirped Residual Phase: Derivation

We now derive the dechirped residual phase expression (Eq. 5) from the frequency definitions and phase-continuity framework established above.

Derivation procedure. Using local time $t' = t - iT$ within the i -th receiver window, the dechirped signal is obtained by multiplying the received chirp (at CFO-scaled parameters β', BW') by the receiver's nominal-parameter conjugate reference chirp (β, BW). The residual phase is the integral of the instantaneous frequency difference.

For an upchirp with integer cyclic shift c and direction v (+1 for upchirp, -1 for downchirp), the transmitted chirp sweeps at rate $\beta'v$ while the reference sweeps at rate βv . Their difference produces:

- A **chirp-rate mismatch** term $(\beta' - \beta)v$, which integrates to $\pi(\beta' - \beta)v \cdot t'^2$.
- A **dechirped center frequency** from the sum of all constant-frequency offsets: f_{cfo} (carrier offset), $BW'(c/N - v/2)$ (code-dependent start frequency under CFO), $\beta'\tau_i v$ (progressive time-offset drift from Eq. 2), and $BW \cdot v/2$ (receiver reference offset). Together these give the linear coefficient.
- The **frame phase** $\phi_i = \phi_0 + 2\pi f_{cfo} \cdot i \cdot T'$ (Section A.3), which enters as the constant term representing the accumulated phase at the start of the i -th symbol.

Combining these three contributions yields the complete residual phase of Eq. 5:

$$\begin{aligned} \Phi_{\text{res}}(t') &= \underbrace{\pi(\beta' - \beta)v}_{\text{chirp-rate mismatch}} t'^2 \\ &+ 2\pi \underbrace{\left(f_{cfo} + BW' \left(\frac{c}{N} - \frac{v}{2} \right) + \beta'\tau_i v + \frac{BW \cdot v}{2} \right)}_{\text{dechirped center frequency}} t' + \underbrace{\phi_i}_{\text{frame phase}} \end{aligned} \quad (36)$$

Specialization to downchirps. For SFD downchirps ($v = -1$), the chirp-rate mismatch reverses sign to $-\pi(\beta' - \beta)$, while the dechirped center frequency becomes $f_{cfo} + BW'/2 + \beta'\tau_i - BW/2$. The magnitude of the chirp-rate mismatch remains the same as for upchirps.

Sync-word upchirps. The sync-word symbols are ordinary coded upchirps with specific cyclic shifts c'_1, c'_2 determined by the LoRa network identifier. Their residual phase follows the same expression with $c = c'_1$ or c'_2 , modifying only the dechirped center frequency (through the $BW'(c/N - 1/2)$ term) while the chirp-rate mismatch and frame phase remain unchanged.

Key universal property. Across all symbol types, the frame phase $\phi_i = \phi_0 + 2\pi f_{cfo} \cdot i \cdot T'$ is a universal signature of CFO, independent of symbol content or type. The chirp-rate mismatch and the dechirped center frequency vary by

symbol type and cyclic shift, but within each symbol type the mismatch is constant; its effect on the FFT peak is therefore a fixed complex multiplier that does not alter the inter-symbol phase progression determined by the frame phase.

A.5 Single-Bin Energy Concentration

The dechirped residual phase (Eq. 5) is quadratic in t' . In principle, the FFT of a signal with quadratic phase involves the Fresnel integral, which has no closed-form solution. We show that the chirp-rate mismatch coefficient is small enough that signal energy remains concentrated in the peak FFT bin, and that the quadratic term contributes a deterministic, symbol-type-constant phase factor that the detection algorithm absorbs.

Quadratic phase bound. The chirp-rate mismatch coefficient is $\pi(\beta' - \beta)$. Using the exact shared-clock relation $\beta' = \beta(1 + \epsilon)^2$:

$$\pi(\beta' - \beta) = \pi\beta(2\epsilon + \epsilon^2) \approx 2\pi\beta\epsilon. \quad (37)$$

The total quadratic phase accumulated over one symbol duration T is:

$$\pi(\beta' - \beta) \cdot T^2 = 2\pi\beta\epsilon \cdot T^2 = 2\pi \frac{BW}{T} \epsilon \cdot T^2 = 2\pi\epsilon \cdot BW \cdot T = 2\pi\epsilon \cdot 2^{SF} \quad (38)$$

For a carrier at f_c with CFO f_{cfo} , $\epsilon = f_{cfo}/f_c$. Typically, $\pi(\beta' - \beta)T^2 < 0.5$ rad (e.g. in worst-case 2.4 GHz, CFO can reach 40 kHz, giving 0.43 rad at SF12), well below 2π . The quadratic modulation therefore causes only minor energy spreading across FFT bins; nearly all signal power remains concentrated in the peak bin k^* .

The same bound applies to downchirps, since the magnitude of the chirp-rate mismatch is identical (only the sign flips).

Consequence: deterministic phase factor and scalloping phase. For sampling rate F_s and N_s samples per symbol, the FFT output at the peak bin can be written in terms of sample index n as:

$$Y_i[k^*] = e^{j\phi_i} \sum_{n=0}^{N_s-1} e^{j(\pi(\beta' - \beta)n^2/F_s^2 + \delta_i n)} \quad (39)$$

where δ_i is the fractional-bin offset (the angular frequency mismatch between the true tone and the nearest bin). The sum $G(\delta_i) = \sum_n e^{j(\pi(\beta' - \beta)n^2/F_s^2 + \delta_i n)}$ is a discrete Fresnel-like factor whose magnitude and phase are determined by the chirp-rate mismatch and δ_i .

Scalloping phase from fractional-bin offset. Even when the chirp-rate mismatch is zero, evaluating the DFT at k^* instead of the true frequency introduces a scalloping phase. For a pure tone at fractional offset δ_i (in radians per

sample), the DFT at k^* is a Dirichlet kernel:

$$G_0(\delta_i) = \sum_{n=0}^{N_s-1} e^{j\delta_i n} = \frac{\sin(N_s\delta_i/2)}{\sin(\delta_i/2)} \cdot e^{j\delta_i(N_s-1)/2} \quad (40)$$

The magnitude exhibits the familiar sinc-like scalloping loss, and the phase is:

$$\angle G_0(\delta_i) = \frac{(N_s-1)}{2} \delta_i \approx \frac{N_s}{2} \delta_i = \pi \tilde{\delta}_i \quad (41)$$

where $\tilde{\delta}_i = N_s\delta_i/(2\pi)$ is the fractional-bin offset in bins. Thus the phase at the peak bin is:

$$\angle Y_i[k^*] \approx \phi_i + \pi \tilde{\delta}_i + \angle G_{\text{quad}} \quad (42)$$

where $\angle G_{\text{quad}}$ is the additional (small, constant) contribution from the quadratic term.

Linearity of the scalloping phase across symbols. For preamble symbols ($c = 0$), the dechirped center frequency drifts linearly due to the progressive time offset $\tau_i = \tau_0 - i\epsilon T$. The bin position k^* typically does not change (total drift < 1 bin over K symbols), so $\tilde{\delta}_i$ inherits the same linear drift:

$$\tilde{\delta}_i \approx \tilde{\delta}_0 + i \cdot \epsilon \cdot 2^{SF} \cdot \frac{\beta'}{\beta} \quad (43)$$

The scalloping phase $\pi \tilde{\delta}_i$ therefore adds a linear contribution to the inter-symbol phase progression, yielding:

$$\angle Y_i[k^*] \approx (\phi_0 + \psi_0) + (2\pi f_{cfo}T' + \psi_1) i \quad (44)$$

where $\psi_0 = \pi \tilde{\delta}_0 + \angle G_{\text{quad}}$ is a constant offset and $\psi_1 = \pi\epsilon \cdot 2^{SF} \cdot \beta'/\beta$ is a small slope correction. Both are deterministic functions of ϵ and τ_0 . The column-wise FFT (Section 4) captures the total slope $2\pi f_{cfo}T' + \psi_1$; since $\psi_1 \ll 2\pi f_{cfo}T'$, the measured phase rotation rate remains an accurate proxy for f_{cfo} .

For SFD downchirps, the sign-flipped chirp-rate mismatch produces a different constant factor, and the reversed dechirped center frequency yields a different scalloping offset. The resulting phase difference between upchirp and downchirp peaks is deterministic and absorbed by the algorithm, which processes these symbol types separately.

Inter-symbol bin stability. For preamble symbols ($c = 0$), the peak bin shifts from symbol to symbol due to the progressive time drift τ_i . The bin shift over K symbols is:

$$\Delta k = \frac{\beta' \cdot K\epsilon T}{BW/N} = K\epsilon \cdot 2^{SF} \cdot \frac{\beta'}{\beta} \approx K\epsilon \cdot 2^{SF} \quad (45)$$

At $K = 8$, $\epsilon = 1.7 \times 10^{-5}$, and $SF = 12$, this gives $\Delta k \approx 0.56$ bins. Thus the signal remains concentrated in a single column of the $K \times N$ spectral matrix, ensuring that the column-wise FFT in the detection algorithm (Section 4) captures the full inter-symbol phase evolution.

A.6 Phase-Coherent Accumulation: SNR Analysis

With the linear-phase approximation established, we formally analyze the SNR gain of coherent versus non-coherent accumulation.

Signal model at the peak bin. At bin k^* , the FFT output for the i -th preamble symbol is:

$$Y_i[k^*] = s \cdot e^{j\phi_i} + n_i \quad (46)$$

where $s = N_s$ is the signal amplitude (real, positive) and $n_i \sim \mathcal{CN}(0, \sigma^2)$ is complex Gaussian noise.

Coherent addition (with known phase). If the phases ϕ_i are known and compensated:

$$Z_{\text{coh}} = \sum_{i=0}^{K-1} Y_i[k^*] \cdot e^{-j\phi_i} = Ks + \sum_{i=0}^{K-1} n_i e^{-j\phi_i} \quad (47)$$

The signal amplitude is Ks ; the noise power is $\mathbb{E}[|\sum n_i e^{-j\phi_i}|^2] = K\sigma^2$ (since the rotated noise remains i.i.d.); the SNR is $(Ks)^2 / (K\sigma^2) = K \cdot s^2 / \sigma^2 = K \cdot \text{SNR}_1$.

B DC-Centered Signed Frequency Extraction

After dechirping a preamble symbol (code $c = 0$) with the nominal downchirp reference, the residual signal is approximately a complex tone at frequency $f_{dc} = f_{cfo} + \beta\tau_0$. When $f_{cfo} < 0$ and τ_0 is small, f_{dc} can be *negative*. In a standard N_s -point DFT, negative frequencies map to high-index bins near N_s , far beyond the signal bandwidth region of the first N bins. Simply truncating the FFT output to the first N bins therefore misses negative-frequency peaks entirely.

To capture both positive and negative dechirped frequencies, we extract a BW -wide window centered at DC from the full N_s -point FFT. The key idea is to concatenate the last $N/2$ bins (which represent negative frequencies $[-BW/2, 0)$) with the first $N/2$ bins (positive frequencies $[0, BW/2)$):

$$\tilde{X}[\ell] = \begin{cases} X[N_s - \frac{N}{2} + \ell] & \ell = 0, 1, \dots, \frac{N}{2} - 1 \quad (\text{negative freqs}) \\ X[\ell - \frac{N}{2}] & \ell = \frac{N}{2}, \dots, N - 1 \quad (\text{positive freqs}) \end{cases} \quad (48)$$

This produces N bins spanning $[-BW/2, BW/2)$ with resolution BW/N . The signed frequency corresponding to bin ℓ is $f(\ell) = (\ell - N/2) \cdot BW/N$.

For the K preamble upchirps, the peak at f_{dc} is consistent across all symbols (since preamble code is $c = 0$ and the inter-symbol bin drift is negligible—see Section A.5). We average the magnitude spectra:

$$\bar{X}_\uparrow[\ell] = \sum_{i=0}^{K-1} |\tilde{X}_i[\ell]| \quad (49)$$

and identify $\ell_\uparrow^* = \arg \max_\ell \bar{X}_\uparrow[\ell]$. The signed upchirp dechirped frequency is:

$$f_\uparrow = \left(\ell_\uparrow^* - \frac{N}{2} \right) \cdot \frac{BW}{N} = f_{cfo} + \beta\tau_0 \quad (50)$$

Similarly, for the 2 SFD downchirps (dechirped with an upchirp reference), the peak is at $f_{dn} = f_{cfo} - \beta\tau_0$:

$$f_\downarrow = \left(\ell_\downarrow^* - \frac{N}{2} \right) \cdot \frac{BW}{N} = f_{cfo} - \beta\tau_0 \quad (51)$$

These signed frequency measurements feed directly into the coarse-fine CFO fusion described in Section 4.

C Algorithm Pseudocode

Algorithm 1 summarizes the complete DeRa detection and decoding pipeline described in Section 4.

Algorithm 1 DeRa: Phase-Coherent Detection and Decoding**Require:** Received signal $r(t)$; spreading factor SF ; column-FFT length M ; sync-word codes c'_1, c'_2 ; candidate-list length $L = 20$ **Ensure:** Decoded payload symbols $\{\hat{c}_i\}$ or detection failure– **Precomputation: High-Resolution Dechirped FFTs** –1: **for** each of the $K+5$ symbol windows (8 preamble, 2 sync, 3 SFD) **do**2: Dechirp with the appropriate reference chirp and compute N_{fft} -point FFT3: **end for**4: Extract DC-centered signed spectra $\tilde{X}^\uparrow, \tilde{X}^\downarrow$

▸ Appendix B

5: $f_\uparrow \leftarrow \text{peak}(\sum_i |\tilde{X}_i^\uparrow|)$; $f_\downarrow \leftarrow \text{peak}(\sum_i |\tilde{X}_i^\downarrow|)$

▸ Eq. 16

– **Column-Wise FFT (Section 4)** –6: Build $K \times N$ spectral matrix Y from N -point preamble FFTs7: $k^* \leftarrow \arg \max_k \sum_i |Y_i[k]|^2$

▸ Peak column

8: $Z[m] \leftarrow \text{FFT}_M[\{Y_i[k^*]\}_{i=0}^{K-1}]$

▸ Zero-padded column-wise FFT

9: $m^* \leftarrow \arg \max_m |Z[m]|$ 10: $\hat{f}_{cfo}^{\text{fine}} \leftarrow m_{\text{eff}}/(M \cdot T)$ ▸ Eq. 14; resolves $f_{cfo} \bmod BW/N$ – **Coarse-Fine Fusion (Section 4)** –11: $\hat{f}_{cfo}^{\text{coarse}} \leftarrow (f_\uparrow + f_\downarrow)/2$ ▸ Unaliased, resolution $BW/(NP)$ 12: $\hat{f}_{cfo} \leftarrow \hat{f}_{cfo}^{\text{fine}} + \text{round}\left(\frac{\hat{f}_{cfo}^{\text{coarse}} - \hat{f}_{cfo}^{\text{fine}}}{BW/N}\right) \cdot \frac{BW}{N}$

▸ Eq. 18

13: $\hat{\tau}_0 \leftarrow (f_\uparrow - \hat{f}_{cfo})/\beta$ – **Coherent Verification and Local Refinement (Section 4.2)** –14: Retain the 20 best $(\hat{f}_{cfo}, \hat{\tau}_0)$ candidates15: **for** each retained candidate (c, m) **do**16: Evaluate (F_1, P_1, F_2, P_2) and gather V_1, V_2 from the precomputed spectra

▸ Eqs. 10–11

17: $S[c, m] \leftarrow \sum_{\text{all sections}} V_{c,m} e^{-j(P_{c,m} + 2\pi\delta f_{c,m} t_{\text{mid}})}$ 18: Detection score $\leftarrow |S[c, m]|/N_s$ 19: **end for**

20: Refine each of the 20 candidates over five CFO and five timing offsets

21: Interpolate each power maximum to refine $(\hat{f}_{cfo}, \hat{\tau}_0)$ 22: Verify each candidate against sync-word symbols c'_1, c'_2

23: Rerank by refined score; retain the top five for decoding fallback

Algorithm 1 DeRa: Phase-Coherent Detection and Decoding (continued)

– **Unified Detection–Decoding Loop (Section 4.4)** –

24: **for** each candidate $(\hat{f}_{cfo}^{(\ell)}, \hat{t}_0^{(\ell)})$ in ranked order **do** ▷ Stage 3: bounded iteration

Stage 1: Phase-compensated magnitude decode

25: Remove CFO: $s_{\text{comp}}(t) = s(t) \cdot e^{-j2\pi\hat{f}_{cfo}^{(\ell)}t}$

26: **for** each payload symbol i **do**

27: Dechirp with corrected timing $T' = T(1 - \hat{\epsilon})$; compute FFT

28: $\Delta\phi_{\text{cfo}}[c] \leftarrow 2\pi\hat{f}_{cfo} \cdot T'(1 - c/N)$ ▷ Eq. 20

29: $\Delta\phi_{\text{TO}}[c] \leftarrow 2\pi \cdot BW \cdot (T' - T)(1 - c/N + N_{\text{pre}} + 4.25 + i)$ ▷ Eq. 21

30: $Y_i^{\text{coh}}[c] \leftarrow V_1[c] + V_2[c] \cdot e^{-j(\Delta\phi_{\text{cfo}}[c] + \Delta\phi_{\text{TO}}[c])}$ ▷ Eq. 22

31: $\hat{c}_i \leftarrow \arg \max_c |Y_i^{\text{coh}}[c]|$ ▷ Eq. 23

32: **end for**

33: **if** CRC($\{\hat{c}_i\}$) passes **then**

34: **return** $\{\hat{c}_i\}$ ▷ Successful magnitude decode

35: **end if**

Stage 2: Phase-coherent decode (if CRC fails)

36: $\hat{\phi}_0 \leftarrow \angle(\sum_i Y_i[\hat{c}_i])$ ▷ ML estimate of initial phase

37: **for** each payload symbol i **do**

38: $\hat{\phi}_i \leftarrow \hat{\phi}_0 + 2\pi\hat{f}_{cfo}(N_{\text{pre}} + 4.25 + i) \cdot T'$ ▷ Eq. 7

39: $\hat{c}_i \leftarrow \arg \max_k \text{Re}(Y_i[k] \cdot e^{-j\hat{\phi}_i})$

40: **end for**

41: **if** CRC($\{\hat{c}_i\}$) passes **then**

42: **return** $\{\hat{c}_i\}$ ▷ Successful phase-coherent decode

43: **end if**

44: **end for**

45: **return** detection failure ▷ All L candidates exhausted
